Do Now · retrieval

Lesson 146 · Network Threats: Attack Paths and Consequences

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

When packets arrive out of order, what can TCP do?

Q2

Which layer in TCP/IP handles IP routing?

Q3

What data structure follows FIFO?

Q4

What is the result of an AND gate with inputs 1 and 0?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Entry point — Suspicious link, exposed service or weak account.

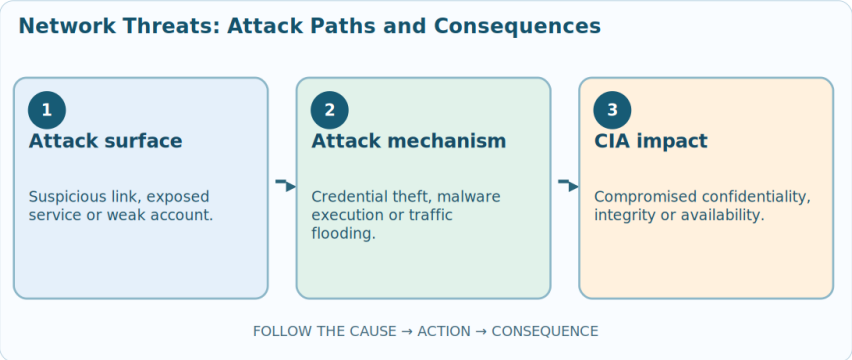
Core idea 2

Threat action — Credential theft, malware execution or traffic flooding.

Core idea 3

Effect on service — Compromised confidentiality, integrity or availability.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

phishing — Social engineering to obtain information or actions through deception.

malware — Software designed to perform harmful or unauthorised actions.

DDoS — Distributed denial-of-service intended to disrupt availability.

CIA triad — Confidentiality, integrity and availability.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A pupil receives a fake school login link and submits a password. Explain the attack.

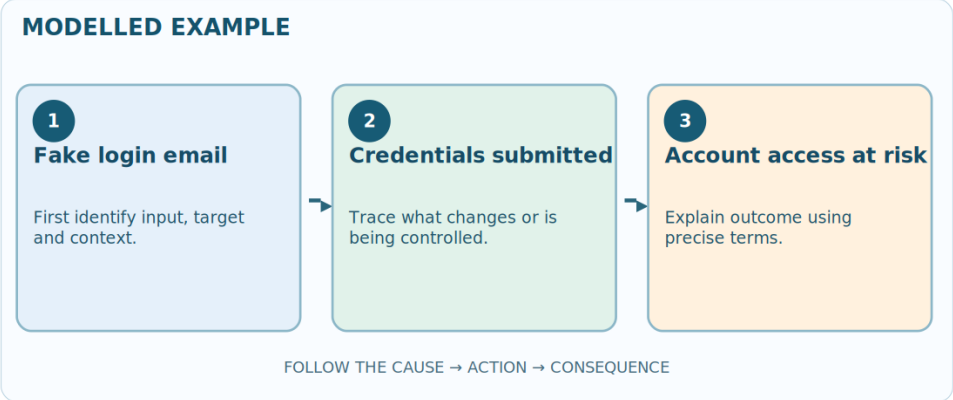
Follow the worked method

1. Fake login email — Identify the facts and make an initial prediction.

2. Credentials submitted — Explain the mechanism or trace the intermediate stage.

3. Account access at risk — Phishing tricks the pupil into visiting an attacker-controlled form. The secret is disclosed, then the attacker may attempt unauthorised account access. Awareness and phishing-resistant MFA help, while a firewall alone does not prevent users entering secrets into fake sites.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Botnet generates traffic
Identify the facts and make an initial prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain how phishing differs from malware and give one example of each. [6]

Question 2

A school site is reachable but pupils' account details are exposed. Identify the affected security property and two controls. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A network suffers DDoS and suspected password theft simultaneously. Prioritise two distinct mitigations and justify why a single firewall rule is inadequate. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Encrypting a site with HTTPS is sufficient to prevent a DDoS attack.

Use key vocabulary from memory

1. Explain phishing in your own words.

2. Explain malware in your own words.

3. Explain DDoS in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.